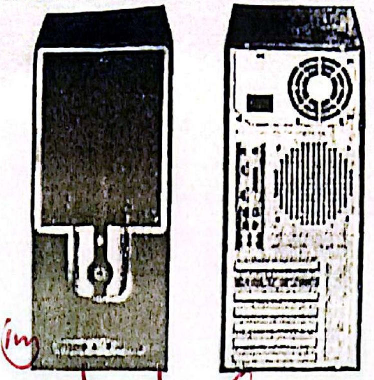
5

(a) Give any two components found on a computer motherboard. (2 marks)

(i) Processor, Bios RAM slot should be in capital letters.

(ii) C mos Battery: Capacitors, transistors Parts. video Cards.

(b) State two major functions of the device given in the image below. (2 marks)



(i) Protecting internal Components housing →

(ii) Earthing → (Protects the user against electric signals. Facilitates the movement of the system as a single unit.

(c) State any one use of ICT in entertainment. (1 mark)

playing Cm. of games, Movies - accessing watching movies - sharing information. Recording live streaming. - editing

2. (a) State any two devices that can be used in a banking sector rather than a Computer. (2 marks)

(i) Point of sales: Biometrics

(ii) Mobile phone, ATM.

(b) Suggest an area where the camera can be used as a solution to society problems. (2 marks)

mention an area scare, mention gadgets and purpose)

Home - hospitals - factories

(i) Can be used by Police (Security

(ii) Can be used on gaming entertainment when far.

(c) Name any one ICT service that may be offered in rural areas. (1 mark)

Mobile banking, Internet service. - E-registration. Printing, Photocopying, E-learning. Paying of bills → betting

3. (a) Distinguish between updating and upgrading of a software on a computer. (2 marks)

→ Changes to existing (i) Typically release Certain enhancement minor changes.

→ replacing to a new (ii) a newer version Major.

(b) Give two reasons why an operating system may be updated. (2 marks)

Hardware Compatibility support.

(i) Add features to provide new support → Improve on user interface

(ii) Provide fixes to possible bugs → Better Performance → Support from development

(c) State any one error that may occur during a booting process. (1 mark)

Hardware failure → Corrupt or missing file

→ Boot device error. → Operating system missing

→ [illegible] → Fan error. → failure to recognise peripherals

→ Failure to recognise the boot drive. → BIOS messages

4. (a) A file stored on a hard disk has 400 KB. When it was sent to me, it was only 60 KB.

Suggest two factors of what may have happened. (2 marks)

- ~~low resolution~~

(i) It may have been Compressed → change file format

(ii) It may have been affected by Virus. → file shortcut

It may have been lost. →

(b) State the term that may be used to group together scattered data on a hard disk and place it in a single place. (1 mark)

Disk Defragmentation.

(c) Give any two advantages of defragmentation on a computer system. (2 marks)

(i) → Programs run faster and efficient → Reduces data redundancy.

(ii) - Increases storage efficiency.

- Reduces space wastage →

Remove if alomer uses To. cinarosou. Wo. Improve

5. (a) You are given the location of a file as follows: (3 marks)

D:\class\drive\application.txt

(i) Identify the drive where the file application is stored.

D. considered D.

(ii) Name any sub-directory in text expression.

- Drive

(iii) Name the term used to represent the whole text expression showing the location of the file.

FILE PATH. without the word file its wrong.

(b) Give the name of the file for the given file extension. (2 marks)

(i) GIF Image / picture / graphics / ~~Animation~~

(ii) MP3 Audio file

6. (a) Name two types of processors that may be found in a computer. (2 marks)

(i) RISC, CISC, Duo Core, AMD, Pentium INTEL, →Celeron.

(ii) ~~Xeon~~, Intel 3, 9, Pentium 2, 4 ~~Xeon~~.

- purpose of the processor → Server Processors, embedded Processor.

(b) State two factors that may affect the processor's performance. (2 marks)

nature of work

Processor architecture

(i) Clock speed, Cache size Hard disk Size

(ii) ~~Ram size~~, Size of Ram. Capacity ~~size~~

Register size

✓ type of Operating system.

✓ Over heating of the processor

Age of the processor

(c) State the type of memory that is embedded within a microprocessor. (1 mark)

Cache memory.

7. (a) A student has a flash disk of 3600 MB. He would like to copy several files from the hard disk totaling to 6 GB.

Calculate the extra space he will need to copy all the files. (4 marks)

Flash size = 3600 MB

Size of all files 6gb × ~~1024~~ = 3000 MB

Extra space required 60~~44~~ - 3600 = 2~~5~~40MB.

1 GB = 1024m

Y = 3600MB

$1024 = \frac{3600}{1024}$

= 3.52

6000 - 3~~52~~

= 2.48GB

6000 - 3600 = 2.48GB.

(b) Give an example of a storage media that uses light during writing. (1 mark)

DVD , Compact dis (CDs)
BLU-Ray

8. (a) A microprocessor running at 2 clock cycles per instruction and the number of instructions per second is 10^6. Determine the clock speed in Hertz. (2 marks)

No: of instruction per sec = 1,000,000

Clock speed = CPI × IPS. $=(10^6 \times 2)$ Hz

= 1,000,000 × 2   2,000,000Hz ✓

(b) Define an instruction cycle. (2 marks)

Sequence of operation that a processor has to carry out while executing the instruction

(c) Use the given list below to select the smallest unit of data which can be stored; Kilobyte, Byte, Bit, MB. (1 mark)

Bit

9. (a) State the type of software that is designed to control the operation of a computer. (1 mark)

System Software

(b) Give an example of: (2 marks)

(i) A low level language Assembly, Machine language.

(ii) Utility program Antivirus, Disk cleanup, Disk defragmenter, file compression

(c) Name the two types of Operating System interfaces which are available to a user. (2 marks)

(i) Graphic user interface

(ii) Command line interface

(Standard acronyms in capital letters)

Note Consider any two among them.

10. State TRUE or FALSE in the table corresponding to the given statement (5 marks)

No substitute for true or false

	Statement	TRUE or FALSE
(i)	A program is a set of instruction.	TRUE
(ii)	Cobol is a low-level Language.	FALSE
(iii)	A Process is a program in Execution.	TRUE
(iv)	An Operating system is an Application software.	FALSE
(v)	An Antivirus can create documents.	FALSE
(vi)	Device drivers are managed by the O/S.	TRUE.

Do not accept if a learner ticks.

11. (a) Explain how the following professionals can apply ICT to improve on their service delivery.

(i) Teacher (2 marks)

Use a projector to deliver a lesson in class

Using video conferencing software to organise online lessons

Using a printer to produce hard copies

Using a computer to register learners

Conduct experiments through simulation.

use a computer to assess and evaluate learners

Ste: Store patients records.

(ii) Doctor (2 marks)

Using DNA task kits to determine the Paternity of adult.

Using ICT to carry out medical research

Use of telemedicine apps to remotely contact patients and colleagues

(b) Mention one health issue that may arise due to prolonged use of computers. (1 mark)

→ Obesity. Neck Pain, Back pain, Repetitive Strain injury

→ Stress. wrist Pain Eye strain. carpel tunnel syndrome

12. A secondary school has reported several issues in its computer laboratory, including slow booting of computers, accumulation of dust inside the machines, unlabeled power cables, and regular student misuse of the equipment.
As a senior six ICT student assigned to assist in restoring proper laboratory function;

(a) Outline two preventive maintenance practices you would implement to promote the long term functionality of the computers. (2 marks)

(i) Blowing of dust

(ii) installation of surge

(b) Identify three practical measures to secure the computer laboratory. (3 marks)

(i) Proper Cable installation.

(ii) Installation of fire extinguishers

(iii) Installation of lightning Conductors.

Use of standard furniture. Regular cleaning of the labs

13. (a) Explain the meaning of the term user interface. (2 marks)

refers to the point of interaction between the the user and the Computer to enable Communication.

(b) Describe briefly the use of the following Graphical user interface (GUI) features.

(i) Check box (1 mark)

Allows the user to select one or more options from a set

(ii) Radio button (1 mark)

select only one item at a time.

(iii) Text boxes (1 mark)

Allows user to enter text.

14. A company is planning to automate its business processes using appropriate software. As an ICT consultant, advise the company by:

(a) Recommending **two** types of application software that would be suitable for managing sales and employee records. (2 marks)

(i) Database management Software Access

(ii) Spreadsheets Excell x

Word processing software ~~Ms word~~ x | word processors

(b) How would the above software improve the company's efficiency. (3 marks)

- Providing updated information.
- Auto-calculation and data analysis

(i) Easy sharing of information Data security

(ii) Reduce delays , Data recovery → Reduces data redundancy

Automated changes: reduces on the errors.

15. (a) Differentiate between a web browser and a web server. (2 marks)

Web browser is a program used to allow users access remove and view documents on the Internet / webpages

a web server is a dedicated Computer that stores , distributes and manages web pages / resources

Conjunction is require

(b) Given the following e-mail address:
info@bbs.co.ug
What is the meaning of the following parts?

(i) Info (1 mark)

User name / account holder

(ii) bbs.co.ug (1 mark)

Domain Name / Text version of the Internet Protocol Address.

(c) Give **one** example of a web browser that can be used to read mails sent to the above address. (1 mark)

- Netscape Navigator - Google chrome , Mozilla fire: Safari opera mini. Microsoft Edge

16. (a) A secondary school in a rural area intends to introduce e-learning by connecting the school to the internet. As the ICT advisor outline **three** key considerations the school must take into account when choosing an Internet Service Provider (ISP). (3 marks)

(i) Coverage Cost Software Compatibility

(ii) Speed Reliability Scalability

(iii) Bandwidth experience reputation.

(c) Suggest **two** advantages of cloud computing. (2 marks)

(i) Provides unlimited storage sources

(ii) Provides Security to data Automatic updates

- Provides Back up. It enables easy sharing of information.

~~Provides Security to data~~

17. (a) Explain the meaning of the term "Virtual reality" (1 mark)

is where Computers are used to Create an artificial environment that users Can interact with as if it ~~was~~ is real

(b) Mention any **two** areas where virtual reality is applied. (2 marks)

(i) Allowing architects to walk around

(ii) Training soldier in Combat
Training surgeons
computer games.

(c) State the role of the following careers in the ICT field. (2 marks)

(i) Software engineer.

Develops system and application softwares
- Testing software
Develops user and technical documentation for the new software

(ii) Database administrator.

Designs and manages implementation and maintenance of databases.
Setting up security measures.

18. Computer viruses are one of the security risks that can lead to data loss.

(a) Identify any two devices that can aid transmission of computer viruses from one computer to another. (2 marks)

(i) Flash disks external hard disks.

(ii) Memory, Networking devices
Floppy disks;

(b) Suggest any two ways through which viruses can be controlled. (2 marks)

Avoid sharing files from unknown sources

(i) installing updated antivirus software

(ii) Carrying out full system Scan. · Always updating
Scanning all removable drives before using them

(c) Identify any one other risk that can lead to data loss in an organization. (1 mark)

Denying of unknown people passwords
- Hacking and Cracking
- Trojan attack
Outdated software
Natural disaster
Theft
System failure

19. Use the figure below to answer the questions that follow.

	A	B	C	D	E	F
1	HOUSE	ATHLETICS	FOOTBALL	SWIMMING	TOTAL	POSITION
2	TIGER	12	64	0		
3	LION	21	43	8		
4	SHARK	3	2	300		
5						
6	MAX					

(a) Identify the application program that is appropriate to manipulate this data. (1 mark)

MS Excell Microsoft excell

(b) How many columns do you see in the figure above? (1 mark)

Six Colums.

(c) State the formula that can be used to find the total number of points collected by all houses. (1 mark)

=Sum(E2,E3,E4)=Sum(E2:E4)

= ~~(D2+E2+D2)~~ =E2+E3+E4

(d) What function should be applied to compute the position column. (1 mark)

RANK.

(e) What function would be applied to find the highest points scored in swimming? (1 mark)

MAX

20. (a) A school secretary was given a report to typeset to be handed over to the Headteacher. During the process of typesetting, some words were underlined with a red wavy line while others were underlined with a green wavy line. Help her understand their meaning.

(i) Red wavy underline. (1 mark)

~~Red wo~~ Wrong spelling of the word

(ii) Green wavy underline. (1 mark)

Grammatical errors

(b) Suggest two formatting features of a word processor. (2 marks)

(i) Font Color. line spacing - WordArt.

(ii) Text alignment Font style underlining Italice

(d) Identify one commonly used word processor in schools. (1 mark)

MS WORD. (Microsoft Word

END